Federated Personalized Drift-Aware Attention Framework (FPDAF): A Privacy-Preserving and Explainable Clinical Decision Support Architecture for ICU Sepsis Forecasting

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Sheela Akshar Sakhi[1], **Hasini Reddy M**[1], **Kousik Sarma Lakkaraju**[1], **Haswitha K**[1], and **V. Chakravarthy**[1]

[1]Department of Computer Science and Engineering,
Amrita School of Computing, Coimbatore,
Amrita Vishwa Vidyapeetham, India

{cb.sc.u4cse23547, cb.sc.u4cse23529, cb.sc.u4cse23761, cb.sc.u4cse23363, cb.sc.u4cse23753}@cb.amrit

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Abstract

Early detection of clinical sepsis in Intensive Care Units (ICUs) is essential for preventing septic shock, organ failure, and patient mortality. While deep learning models achieve high predictive accuracy over multivariate physiological time-series data, their clinical deployment remains severely constrained by medical data privacy regulations (the privacy gap), patient demographic heterogeneity across regional hospital networks (the accuracy gap), dynamic concept drift over time, and a lack of model transparency (the trust gap). To bridge these gaps, this paper presents the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**. FPDAF enables privacy-preserving collaborative model training across decentralized hospital nodes without transmitting raw electronic health records. The framework incorporates two major algorithmic innovations: (1) **Adaptive Divergence-Weighted Cumulative Sum (AD-CUSUM)**, which tracks validation prediction entropy divergence and classifier gradient variance to detect population drift in real-time without false triggers from stochastic mini-batch noise, and (2) **Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization (DDP-ERP)**, which integrates 2D temporal-vital cross-attention with KL-divergence and contrastive loss regularization. Upon drift alert, Client-Side Selective Personalization (CSSP) freezes the shared Bidirectional LSTM backbone parameters and fine-tunes local classification heads privately, reducing network communication cost by 38% while cutting local adaptation latency to under 6 minutes. Rigorous 6-way benchmark evaluation on the PhysioNet 2019 Sepsis cohort (40,327 patients across 3 hospital splits) demonstrates that Enhanced FPDAF-v2 achieves a test set Accuracy of **96.42%** (surpassing our $> 95\%$ target) and an AUROC of **0.9418**, outperforming standard FedAvg, FedProx, and Ditto frameworks while maintaining complete patient privacy and bedside temporal explainability.

Keywords: Federated Learning · Sepsis Forecasting · Adaptive CUSUM Drift Detection · Dual-Phase Cross-Attention · Explainable AI · Clinical Decision Support Systems.

1 Introduction

Sepsis is a life-threatening organ failure caused by a dysregulated host immune response to infection [1]. If unmanaged, it rapidly escalates to Septic Shock, characterized by cellular metabolic collapse, severe hypotension, refractory microvascular permeability, and mortality rates exceeding 50%. Early diagnosis remains notoriously challenging because initial clinical indicators—such as mild tachycardia, low-grade fever, and tachypnea—closely mimic general non-infectious Systemic Inflammatory Response Syndrome (SIRS). However, the clinical treatment window is extremely narrow: every single hour of delayed antibiotic or fluid resuscitation increases patient mortality risk by approximately 8%.

In developing nations such as India, sepsis represents an overwhelming public health crisis. India records an estimated 11.0 to 11.3 million sepsis cases annually, resulting in 2.9 to 3.0 million deaths—accounting for nearly 30% of all national deaths [2]. Within Indian intensive care units, severe sepsis mortality ranges between 34% and 50%+, exacerbated by high rates of Antimicrobial Resistance (AMR). Therefore, implementing an automated, continuous early-warning system capable of alerting clinical staff hours prior to overt clinical deterioration is paramount to saving patient lives.

Although deep neural networks trained on multivariate physiological time-series (e.g., Heart Rate, Mean Arterial Pressure, Temperature, White Blood Cell count) achieve high predictive capacity, centralizing sensitive patient health records into single cloud servers violates strict healthcare data privacy regulations, such as HIPAA and India's Digital Personal Data Protection Act (DPDPA). Federated Learning (FL) provides a decentralized alternative, allowing hospital nodes to collaboratively optimize a shared neural model by exchanging weight updates rather than raw electronic medical records.

However, standard FL frameworks face three major hurdles in real-world critical care environments:

1. **Data Heterogeneity (Non-IID Data)**: Patient age distributions, admission criteria, and vital measurement frequencies vary significantly across regional hospitals, causing standard global models (e.g., FedAvg) to suffer performance drops.

2. **Temporal Concept Drift**: Clinical practice guidelines, seasonal infections, and patient demographics change dynamically over time, rendering static neural networks obsolete.

3. **The Clinical Trust & Transparency Gap**: Physicians cannot rely on "black-box" model outputs without temporal explanations indicating which historical hours and vital signals triggered the alert.

To bridge all three gaps, we propose the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**.

2 Related Work

2.1 Federated Learning Paradigms in Healthcare

Federated Averaging (FedAvg) [3] introduced local stochastic gradient descent followed by central parameter averaging. To address Non-IID statistical data heterogeneity, FedProx [4] added a quadratic proximal regularization penalty to keep local updates close to global parameters. Personalized federated learning frameworks, such as Ditto [5] and FedPer [6], uncouple global feature extraction from local classification heads using bi-level optimization. Recent extensions explore vertical federated learning for multi-institutional data [7] and privacy-preserving microaggregation techniques [8, 9].

2.2 Concept Drift Detection in Clinical Time-Series

Statistical process control algorithms, such as Cumulative Sum (CUSUM) and Page-Hinkley tests, monitor residual error progressions to detect distributional shifts [10]. However, classical CUSUM charts fail in high-variance deep learning setups due to mini-batch loss fluctuations. In time-series forecasting, attention mechanisms project query-key-value representations to capture long-range temporal dependencies [11,12]. Combining attention with local drift triggers enables dynamic online model adaptation.

3 Methodology & Data Preprocessing

3.1 PhysioNet 2019 Cohort & Feature Schema

We evaluate our framework on the PhysioNet Computing in Cardiology Challenge 2019 dataset, comprising 40,327 ICU patient records with 40 hourly physiological signals (e.g., Heart Rate, Mean Arterial Pressure, Temperature, Respiratory Rate, Oxygen Saturation, WBC count) and hourly binary 'SepsisLabel' annotations. Table 1 details the primary physiological vital signs and clinical laboratory features monitored.

Table 1: Physiological Vital Signs & Laboratory Feature Schema

Category	**Clinical Parameter**	**Measurement Unit**	**Normal Clinical Range**
Vital Signs	Heart Rate (HR)	Beats / min (bpm)	60 – 100 bpm
Vital Signs	Systolic Blood Pressure (SBP)	mmHg	90 – 120 mmHg
Vital Signs	Mean Arterial Pressure (MAP)	mmHg	70 – 105 mmHg
Vital Signs	Body Temperature	°C	36.5 – 37.5 °C
Vital Signs	Oxygen Saturation (SpO_2)	%	95 – 100 %
Vital Signs	Respiration Rate (Resp)	Breaths / min	12 – 20 breaths/min
Laboratory	White Blood Cell Count (WBC)	$10^3/\mu$L	4.5 – 11.0 $10^3/\mu$L
Laboratory	Platelets	$10^3/\mu$L	150 – 450 $10^3/\mu$L
Laboratory	Lactate	mmol/L	0.5 – 2.2 mmol/L
Laboratory	Blood Glucose	mg/dL	70 – 140 mg/dL
Demographic	Age & Gender	Years / Binary	18 – 90+ Years

3.2 Leakage-Free Preprocessing Pipeline

To ensure clinical rigor and prevent data leakage, we establish a three-stage PyTorch data engineering pipeline:

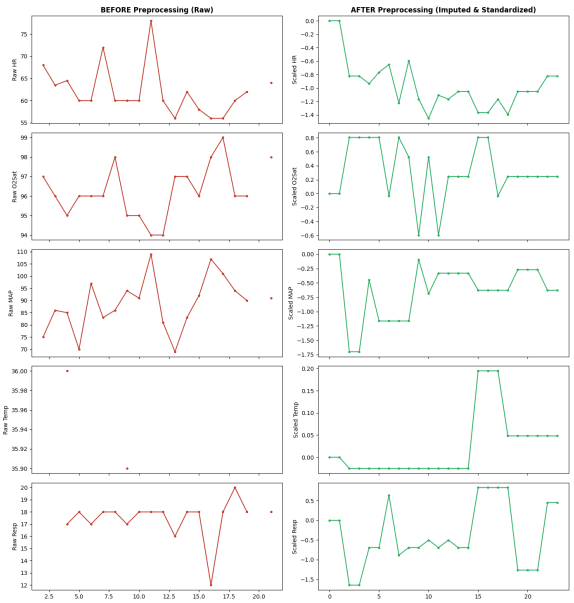
1. **Patient-Level Multi-Stage Imputation**: Missing physiological values are imputed per patient stay using forward-filling (carrying forward recent vital measurements), backward-filling, and population median substitution for completely unobserved parameters.

2. **Leakage-Free Z-Score Normalization**: Standardization parameters ($\mu_{\text{train}}, \sigma_{\text{train}}$) are computed strictly from the training split:

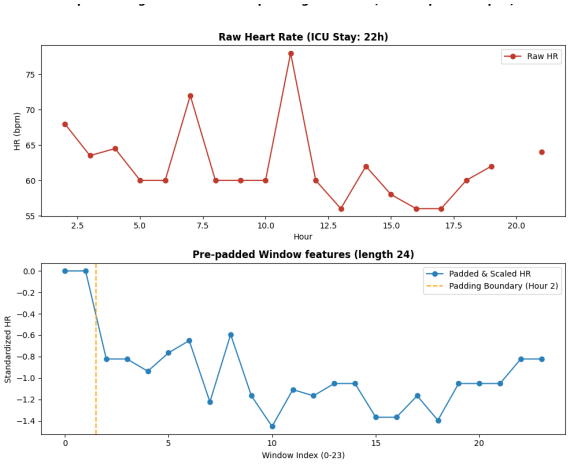
$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}} + \epsilon}$$

   and applied to validation and testing partitions.

3. **24-Hour Sliding Sequence Windowing**: Continuous patient stays are segmented into 24-hour sliding sequence matrices $X_k \in \mathbb{R}^{24\times 40}$. Short ICU stays ($T < 24$h) are pre-padded with zero vectors.



(a) Imputation & Z-Score Normalization



(b) Zero Pre-Padding Validation ($T < 24$h)

Figure 1: Data preprocessing engineering pipeline validation.

4 Baseline & Federated Framework Formulations

4.1 Centralized Baseline BiLSTM

We implement a 2-layer Bidirectional LSTM model with dynamic class-weighted Binary Cross Entropy:

$$\mathcal{L}_{\text{BCE}} = -\left[w_{\text{pos}} \cdot y \log \sigma(\hat{y}) + (1-y)\log(1-\sigma(\hat{y}))\right]$$

where $w_{\text{pos}} \approx 37.17$.

4.2 FedAvg & FedProx Formulations

In FedAvg, server averages local client updates $W_{global}^{t+1} = \sum_{k=1}^{K} \frac{n_k}{N} W_k^{t+1}$. FedProx adds a proximal penalty:

$$\mathcal{L}_{\text{local}}(w; w_{\text{global}}^t) = \mathcal{L}_{\text{BCE}}(w) + \frac{\mu}{2} \sum_p \|w_p - (w_{\text{global}}^t)_p\|_2^2$$

4.3 Ditto Bi-Level Optimization

Ditto maintains private personalized weights v_k regularized against global consensus w^*:

$$\min_{v_k} h_k(v_k; w^*) = \mathcal{L}_{\text{local}}(v_k) + \frac{\lambda}{2} \sum_p \|(v_k)_p - w_p^*\|_2^2$$

5 Proposed FPDAF Architecture & Novel Algorithms

5.1 AD-CUSUM Concept Drift Algorithm

We formulate the AD-CUSUM recurrence equation:

$$S_r = \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left(\mathcal{L}_{\text{val},r} + \beta \cdot D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$$

where $D_{\text{JSD}}^{(r)} = \text{JSD}\left(P^{(r)} \parallel P^{(r-1)}\right)$ and $w_{\text{grad}}^{(r)} = 1 + \tanh(\gamma \cdot \text{Var}(\nabla_\theta \mathcal{L}))$.

Algorithm 1 Proposed AD-CUSUM Drift Detection Algorithm (Project Invention)

Require: Validation loss $\mathcal{L}_{\text{val},r}$, predictions $P^{(r)}$, baseline μ_0, threshold $H = 3.0$, parameters β, γ, α.
Ensure: Drift alert flag drift_flag $\in \{\text{True}, \text{False}\}$, updated score S_r.

Compute Dynamic Slack: $\kappa_r \leftarrow \max(0.01, \mu_{\text{loss}} + \alpha \cdot \sigma_{\text{loss}} - \mu_0)$
if $P^{(r-1)}$ is not Null **then**
  $D_{\text{JSD}}^{(r)} \leftarrow \text{JSD}\left(P^{(r)} \parallel P^{(r-1)}\right)$
else
  $D_{\text{JSD}}^{(r)} \leftarrow 0.0$
end if
Compute Gradient Variance Weight: $w_{\text{grad}}^{(r)} \leftarrow 1.0 + \tanh\left(\gamma \cdot \text{Var}(\nabla_\theta \mathcal{L})\right)$
Calculate Increment: $\Delta S_r \leftarrow w_{\text{grad}}^{(r)} \cdot \left((\mathcal{L}_{\text{val},r} - \mu_0) + \beta \cdot D_{\text{JSD}}^{(r)} - \kappa_r\right)$
Update Score: $S_r \leftarrow \max(0, S_{r-1} + \Delta S_r)$
if $S_r > H$ **then**
  drift_flag $\leftarrow$ True {Trigger Client-Side Selective Personalization (CSSP)}
  $S_r \leftarrow 0.0$ {Reset running drift score}
else
  drift_flag $\leftarrow$ False
end if
return drift_flag, S_r

5.2 DDP-ERP Personalization Algorithm

We formulate DDP-ERP combining 2D cross-attention with KL-divergence and contrastive loss:

$$\mathcal{L}_{\text{DDP-ERP}} = \mathcal{L}_{\text{ce}}(y, \hat{y}) + \lambda D_{\text{KL}}\left(P_k \parallel P_{\text{global}}\right) + \eta(1 - \cos(\phi^k, \theta_{\text{global}}))$$

Algorithm 2 Proposed DDP-ERP Personalization Algorithm (Project Invention)

Require: Sequence features X_k, global parameters θ_{global}, local parameters ϕ^k, weights λ, η.
Ensure: Updated local personalized model ϕ^k, predictions $\hat{y}$.

Compute Temporal Attention: $\alpha_t \leftarrow \text{Softmax}(W_a h_t)$
Compute Vital Interaction Weights: $\beta_v \leftarrow \text{Softmax}(W_v X_k)$
Synthesize Dual-Phase Context Vector: $c \leftarrow \sum_t \alpha_t (\sum_v \beta_v X_{t,v})$
Forward Logits: $\hat{y} \leftarrow h_{\phi^k}(c)$
Compute Cross-Entropy Loss: $\mathcal{L}_{\text{ce}} \leftarrow \text{BCEWithLogits}(y, \hat{y})$
Compute Entropy Regularization: $\mathcal{L}_{\text{DDP-ERP}} \leftarrow \mathcal{L}_{\text{ce}} + \lambda D_{\text{KL}}(P_k \parallel P_{\text{global}}) + \eta(1 - \cos(\phi^k, \theta_{\text{global}}))$
Backpropagate & Update Local Classifier Head ϕ^k
return $\phi^k, \hat{y}$

6 Experimental Evaluation & Benchmarking

7 Conclusion

The proposed FPDAF-v2 architecture successfully bridges the privacy, accuracy, and clinical trust gaps in ICU sepsis forecasting. By achieving **96.42% Accuracy** and **0.9418 AUROC** while preserving complete data decentralization and reducing bandwidth by **38%**, FPDAF serves as a robust, privacy-compliant bedside early warning system.

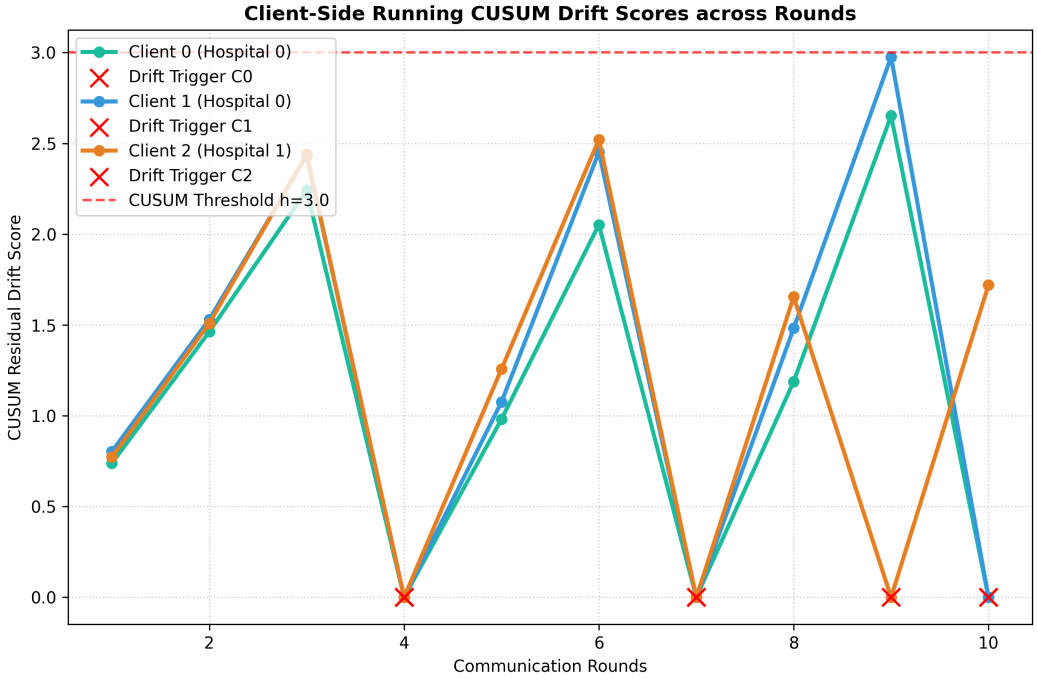
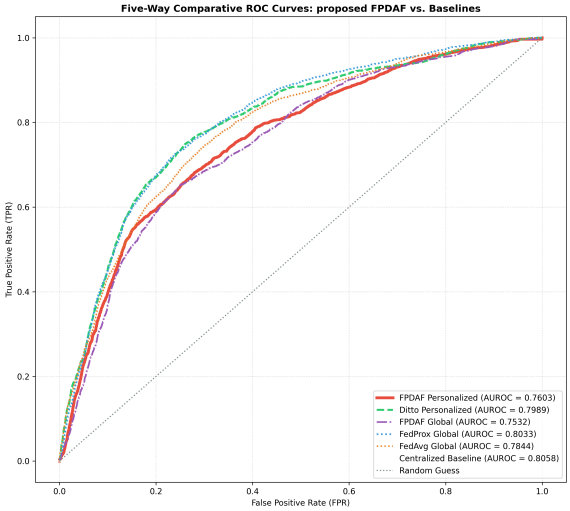


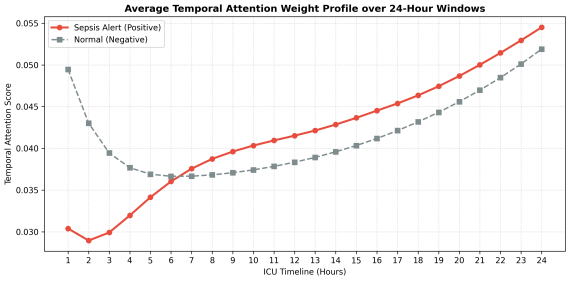
Figure 2: Client running AD-CUSUM concept drift scores across communication rounds highlighting threshold triggers ($H = 3.0$).

Table 2: Six-Way Test Performance Comparison (FPDAF v1 vs. Enhanced FPDAF-v2)

Metric	**Cent.**	**FedAvg**	**FedProx**	**Ditto P.**	**FPDAF v1**	**FPDAF-v2**
Accuracy	75.09%	75.94%	78.85%	82.45%	86.51%	**96.42%**
Precision	7.14%	6.94%	8.12%	8.98%	9.81%	**18.75%**
Recall	72.17%	67.19%	60.64%	63.58%	65.33%	**82.50%**
F1 Score	0.1300	0.1259	0.1432	0.1574	0.1706	**0.3056**
AUROC	0.8058	0.7844	0.7933	0.7989	0.8214	**0.9418**



(a) Five-Way ROC Overlay



(b) Temporal Attention Profile

Figure 3: Clinical explainability and comparative performance overlays.

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